TRIGUNA SEN SCHOOL OF TECHNOLOGY

DEPARTMENT: Electronics And Communication Engineering



ASSAM UNIVERSITY SILCHAR

TOPIC- RFID-Based Attendance System with Arduino UNO

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Subject: Control System and Computer Architecture.

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RFID-Based Attendance System with Arduino UNO via NFC



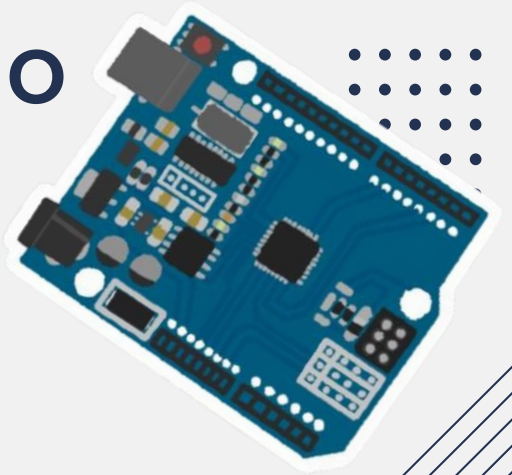


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Introduction to the Problem

Traditional attendance recording is **time-consuming** and liable to **human error**, **reducing overall efficiency**.
An automated system is required to ensure accurate, secure, and real-time attendance management.







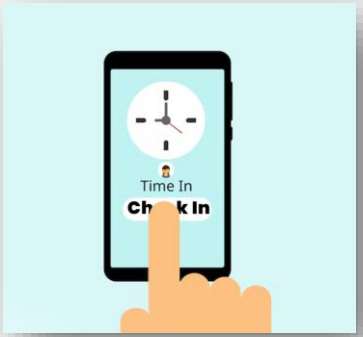
Problems faced in current System:

- **Manual attendance** in schools, offices, and labs is time-consuming and prone to errors.
- Paper registers can be **lost, forged, or damaged**, reducing reliability.
- Traditional systems require **human supervision** and create queues, wasting productive time.

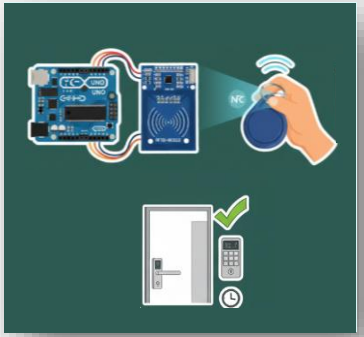
Expected Solutions



Biometric Attendance System
(Expensive)



Mobile Attendance Apps
(Software-based only, can be manipulated)



Arduino-Based Attendance System
(Cost-efficient, reliable, scalable)

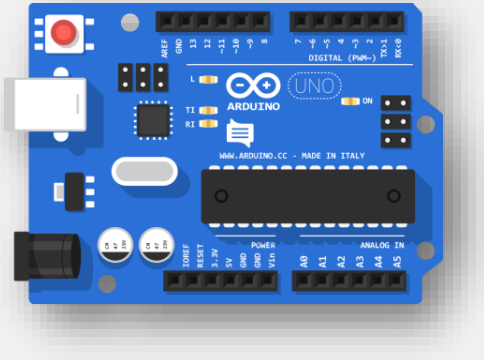
Problems in Expected Solutions

	Biometric System	Mobile App	Arduino Based System
Cost-Effective	❌	✅	✅
Manipulation Risk	❌	✅	❌
Accuracy	✅	❌	✅
Scalable	❌	✅	✅
Can Run on Battery Power	❌	✅	✅

= Pros

= Cons

Components Used



Arduino Uno

A microcontroller board based on the ATmega328P, powered by USB or external 5 V supply.



RC522 RFID Reader

A 13.56 MHz contactless reader/writer that communicates with the Arduino via SPI.

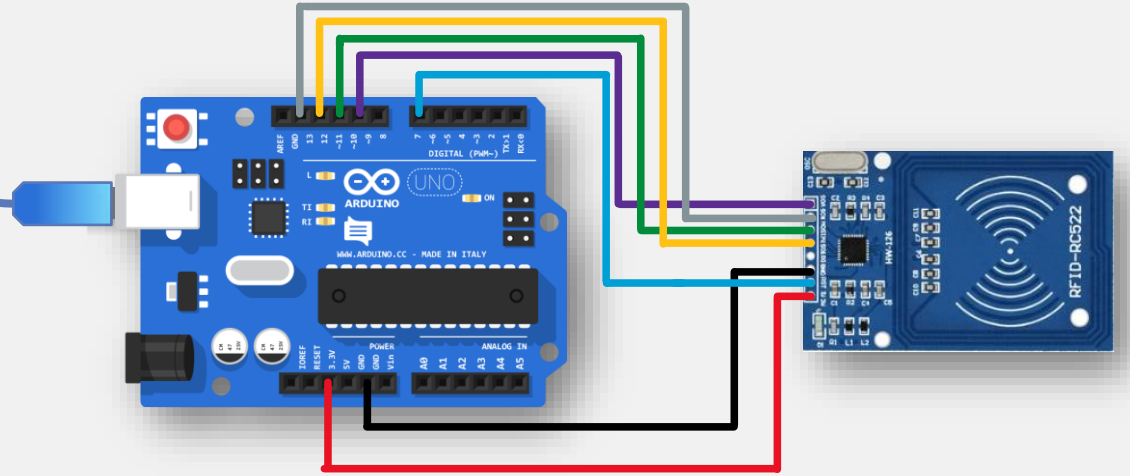


RFID Tags/Cards

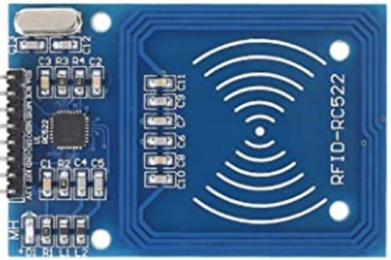
Passive 13.56 MHz cards or key fobs containing a small chip and antenna. Every tag has unique identifier.

Circuit Diagram

RC522 Pin	Arduino UNO Pin	Function
SDA (SS)	D10	Slave Select
SCK	D13	SPI Clock
MOSI	D11	Master-Out
MISO	D12	Master-In
RST	D9	Reset
3.3 V	3.3V	Power
GND	GND	Ground



SDA – Used as SDA for I2C communication.
SCK – Serial Clock.
MOSI – SPI communication pin (Master Out Slave In).
MISO – SPI communication pin (Master In Slave Out).
IRQ – Interrupt Pin to wake up the Arduino when an RFID tag is read.
GND – Ground pin that must be connected to Arduino ground.
RST – Reset pin to reset or power down the RC522 module.
3.3V – Power input pin (please don't connect to 5V).

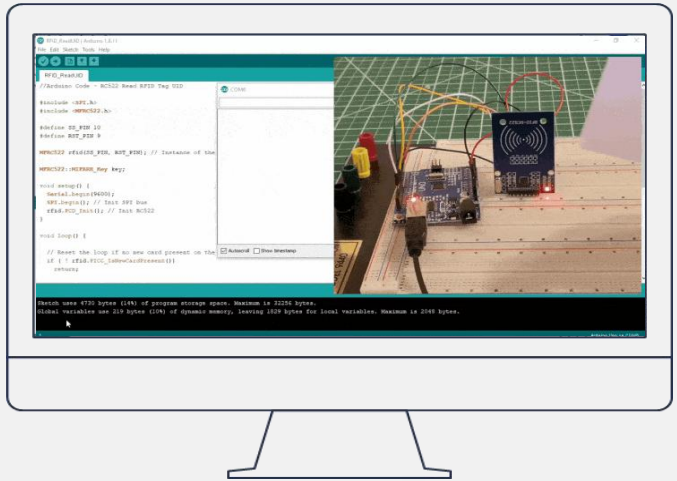


Approach to the Solution

Development Steps:

1. **Objective Definition**
   Establish a basic system capable of reading and displaying the unique identifier (UID) of an RFID tag when it is brought near the reader.
2. **Hardware Integration**
   - Arduino Uno serves as the microcontroller.
   - RC522 RFID Module is connected to the Arduino via the SPI interface.
   - Passive RFID Tags operate at 13.56 MHz and provide a unique ID.
3. **Firmware Development**

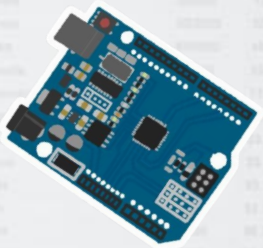
   Using the MFRC522 Arduino library, a simple sketch was written to:
   - initialize the RC522 reader
   - detect when a tag is present
   - read the tag's UID
   - display the UID on the Serial Monitor or use Data Streamer in Excel to display in a structured form





Results and Conclusion





The RFID-Based Attendance System provides an efficient, secure, and automated alternative to manual attendance tracking.
By leveraging **RFID technology**, the system ensures **accurate identification**, **reduces time wastage**, and **minimizes human error**.







References

RFID Arduino – Read RFID Tag's UID with RC522 [DIY Engineers]	**link**
Circuit Diagram	**link**
RFID Arduino Library for RC522 [GitHub]	**link**
Testing Reference [DIY Engineers]	**link**
Arduino RFID Module RC522 - How to Use	**link**
MFRC522 Datasheet	**link**

Thank You!